

String Functions & Character Encoding

String functions are built-in SQL functions that accept character data as input and return manipulated character data

Common Use Cases

Formatting output

```
SELECT UPPER(Name)
FROM Students;
```

Result

```
AADYA
ARPIT
```

Case-insensitive searching

```
SELECT *
FROM Students
WHERE LOWER(Name) = 'aadya';
```

Character Encoding?

Computers store everything as numbers.

Encoding defines

Which number represents which character.

ASCII

Old encoding.

Uses

- 7 bits

Unicode

Designed to represent every language.

Supports

- Hindi

Supports

- 128 characters

Examples

A	65
a	97

Limitation

Cannot represent most world languages.

- Chinese
- Japanese
- Arabic
- Emojis
- Mathematical symbols

UTF-8

Most common encoding today.

Characteristics

- Variable length
- 1–4 bytes per character
- Backward compatible with ASCII
- Used by MySQL, PostgreSQL, web browsers, APIs

Preferred encoding for modern applications.

aadya | AADYA | Aadya

Are these equal?

Answer

Depends on the database collation.

What is a Collation?

A collation determines

- comparison rules
- sorting rules
- case sensitivity

MySQL

Most default collations are case-insensitive.

Example

```
WHERE Name='aadya'
```

may also match

```
AADYA
```

Case-Sensitive Search

Example

```
SELECT *  
FROM Users  
WHERE BINARY Username='Aadya';
```

Returns only

```
Aadya
```

Not

```
AADYA
```

```
SELECT UPPER(NULL);
```

Result

```
NULL
```

Vendor Differences

Oracle

Empty string



is treated as

```
NULL
```

Unicode Considerations

UPPER() works with Unicode-aware databases.

Example

German

straße

may become

STRASSE